

Who Speaks for the Organization? Simulating Strategic Organizational Reasoning in Regulatory Comment Processes

Anita Srinivasan
UC Berkeley School of Law
Berkeley, California, USA
anita@berkeley.edu

Abstract

Large language model (LLM) policy simulations typically model individuals expressing authentic preferences. Yet the actors who most shape regulatory outcomes are organizations producing strategic documents designed to build legal records, not express beliefs. We test whether modeling internal organizational deliberation—decomposing an organization into legal, business, and government affairs departments before synthesizing a public comment—produces more realistic simulated regulatory comments than either individual agents or monolithic organizational agents. Using the SEC’s proposed amendments to Rule 3b-16 (expanding the definition of “exchange” to cover DeFi protocols) as a case study, we simulate comments from eight real organizations across three conditions and validate against the actual filed comment record. We find that monolithic organizational agents achieve near-perfect position prediction (100% accuracy) and high argument alignment, while deliberative agents produce marginally broader arguments but introduce a position error in one case—a community banking association whose internal departments recommended cautious support but whose synthesis agent flipped to opposition. We also apply the same architectures to an anticipated SEC rulemaking with no existing comment record, demonstrating the tool’s forward-looking predictive use case.

Keywords

LLM agents, policy simulation, regulatory comments, organizational reasoning, multi-agent deliberation

1 Introduction

A growing body of work uses LLMs to simulate public participation in policy processes, part of a broader concern about how AI systems mediate and reshape cultural processes [3]. Systems like VacSim model individual vaccine attitudes [1], InsNet-CRAFTY simulates institutional stakeholders in environmental governance [2], Law in Silico simulates legal societies with individual decision-making and institutional mechanisms [12], and recent work on multi-agent negotiation explores strategic interaction between LLM stakeholders [4]. These approaches share a common assumption: that each policy actor can be modeled as a single agent with a coherent position.

In practice, the actors who most shape U.S. regulatory outcomes are *organizations*—trade associations, advocacy groups, and regulated entities—producing *strategic documents* during formal comment periods. The gap between what an organization privately

believes and what it strategically argues in a public comment letter is the core challenge. A cryptocurrency exchange does not simply “oppose” a regulation; its legal team analyzes statutory authority, its business team models competitive impact, and its government affairs team crafts messaging for the regulator. The final comment letter is a synthesized strategic artifact, not an authentic expression of preference.

We ask: does modeling this internal deliberation process produce more realistic simulated regulatory comments? We test three agent architectures—individual, monolithic organizational, and deliberative organizational—against the actual comment record for SEC Rule 3b-16, a proposed expansion of the definition of “exchange” under the Securities Exchange Act that would bring DeFi protocols under securities regulation.

2 The Strategic Communication Gap

We distinguish three types of agent reasoning relevant to policy simulation:

- (1) **Authentic preference** (individual surveys): An agent expresses what it genuinely believes, informed by its demographic and experiential traits. This approach builds on foundational work showing that LLMs can simulate individual economic decision-making [5] and reproduce survey response distributions across demographic subgroups [6].
- (2) **Institutional mandate** (monolithic organization): An agent acts on behalf of an organization with a stated mission and constituency.
- (3) **Strategic production** (deliberative organization): Multiple agents representing internal organizational functions deliberate, and a synthesis agent produces a document optimized for *persuasive effectiveness with the regulatory audience*.

To illustrate why the distinction matters: in the actual comment record for Rule 3b-16, Coinbase’s comment letter frames its opposition primarily around investor protection and the impossibility of DeFi protocol compliance—not around the threat to its business model. This is strategic framing: the organization argues on grounds the SEC is institutionally obligated to care about (investor protection, administrability) rather than grounds that reveal its commercial self-interest. The gap between what an organization believes and what it strategically argues is precisely what a policy simulation must capture to be useful.

Prior work has explored the first two reasoning types. InsNet-CRAFTY models deliberation *between* institutional agents—including policymakers, lobbyists, and operational agencies—but does not decompose a single organization into internal departments that deliberate before producing a unified output [2]. Our Condition C takes this complementary approach: each organization is decomposed

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into functional departments that may reach different conclusions, and a synthesis step must reconcile internal tensions—an architectural choice that, as the ICBA case demonstrates, can produce emergent behavior absent from monolithic designs. Multi-agent negotiation benchmarks model strategic *action selection* but not *rhetorical framing*—the choice of which arguments to make and how to frame them [4]. The Empty Chair uses LLM personas as deliberation aids for humans in citizens’ assemblies, not as simulations of actual organizational strategic behavior [11].

The unique challenge of regulatory comments is that they are designed to *build a legal record*. Organizations argue what they want courts to hear in any future challenge to the rule, not what they privately believe. The SEC’s Rule 3b-16 rulemaking provides an ideal test case: it received over 250 comments dominated by sophisticated organizational actors, with a clear divide between crypto-native firms opposing the rule, consumer advocates supporting it, and traditional financial institutions taking varied positions [7].

3 Experiment Design

3.1 Case Study: SEC Rule 3b-16

In 2022–2023, the SEC proposed amending Exchange Act Rule 3b-16 to expand the definition of “exchange” by replacing the requirement that a system “bring together” orders with the broader concept of making available “trading interest,” and expanding “methods of interaction” to include “communication protocols.” The rule would require DeFi protocols and other systems meeting the new definition to register as national securities exchanges or comply with Regulation ATS [7].

3.2 Three Conditions

Condition A (Individual):

[Persona: demographics, beliefs] --> [Comment]

Condition B (Monolithic Org):

[Org mandate + constituency] --> [Comment]

Condition C (Deliberative Org):

[Legal Dept] ---+

[Business] ---+--> [Synthesis Agent] --> [Comment]

[Gov Affairs] ---+

Figure 1: Three agent architectures for simulating regulatory comments. Conditions A and B use a single agent per comment; Condition C decomposes each organization into three departmental agents whose outputs are reconciled by a General Counsel synthesis agent.

Condition A (Individual Agents): 15 diverse individual agents spanning demographics (ages 22–68, occupations from software engineers to retired teachers, crypto familiarity from expert to none, varied political leanings). Each received a neutral rule summary and wrote a public comment.

Condition B (Monolithic Organizational Agents): 8 agents representing real organizations that commented on the rule: Coinbase, Consensys, Uniswap Labs, SIFMA, Better Markets, ICBA,

Chamber of Commerce, and Blockchain Association. Each received the same rule summary plus an organizational mandate description.

Condition C (Deliberative Organizational Agents): The same 8 organizations, but each went through a three-step deliberation (Figure 1): (1) three departmental agents (Legal, Business Strategy, Government Affairs) independently analyzed the rule from their functional perspective; (2) a General Counsel synthesis agent received all three perspectives and produced the final comment letter.

3.3 Ground Truth and Implementation

We constructed a ground truth dataset from the actual comment letters filed with the SEC, capturing each organization’s position, key arguments, rhetorical strategies, and tone. All surveys were run using EDSL [8] with Claude Sonnet 4 as the underlying model. Each agent received the same neutral 500-word rule summary as stimulus.

3.4 Evaluation Metrics

We evaluate along three dimensions. *Position accuracy* checks whether each agent’s stated position (support/oppose) matches the actual filed comment. *Argumentative range* is scored 0–5 by automated keyword detection across five categories: legal citations, economic arguments, constitutional arguments, technical blockchain arguments, and political/institutional arguments. This metric captures how many distinct categories of argumentation a comment engages with, not the depth or quality of any individual argument. A comment scores one point per category in which at least one marker appears. *Argument alignment* combines position accuracy (0 or 2 points) with thematic overlap between simulated and actual rhetorical strategies (0–3 points, measured by keyword co-occurrence). All scoring is automated and deterministic.

4 Results

4.1 Position Distribution

Table 1: Position distributions across conditions. Ground truth: 6 oppose, 2 support (75% oppose).

Condition	N	Oppose	Support	% Oppose
A (Individuals)	15	7	5	46.7%
B (Monolithic Org)	8	6	2	75.0%
C (Deliberative Org)	8	7	1	87.5%
Ground Truth	8	6	2	75.0%

Condition B achieves a perfect 100% position match with the actual comment record. Condition C matches 7 of 8 positions, misclassifying ICBA—the Independent Community Bankers of America—as opposing when it actually supported the rule. Individual agents (Condition A) produce a more balanced distribution (46.7% oppose), reflecting authentic individual preferences rather than the organizational dynamics that dominated the actual comment record.

Table 2: Argumentative range scores (0–5): number of argument categories present (legal, economic, constitutional, technical, political).

Condition	Average	Min	Max
A (Individuals)	2.87	0	4
B (Monolithic Org)	4.25	4	5
C (Deliberative Org)	4.38	3	5

4.2 Argumentative Range

Both organizational conditions significantly outperform individual agents. Condition C shows a marginal advantage (4.38 vs. 4.25), with its comments more frequently incorporating arguments across multiple categories.

4.3 Organization-Level Alignment

Table 3: Organization-level comparison. Match quality rated 1–5 on position accuracy and argument alignment. Bold indicates a position mismatch.

Organization	GT	B	B Qual	C	C Qual
Coinbase	opp	opp	5	opp	5
Consensys	opp	opp	5	opp	5
Uniswap Labs	opp	opp	5	opp	5
SIFMA	opp	opp	5	opp	5
Better Markets	sup	sup	4	sup	4
ICBA	sup	sup	5	opp	3
Chamber of Comm.	opp	opp	5	opp	4
Blockchain Assn.	opp	opp	4	opp	4
Average			4.8		4.4

4.4 The ICBA Failure: When Deliberation Overrides Identity

The most instructive result is the ICBA case. In Condition B, the monolithic agent correctly identified the organization’s position: community banks support the rule because it levels the competitive playing field between regulated banks and unregulated crypto platforms. In Condition C, all three departmental agents recommended *cautious support*, yet the synthesis agent flipped to opposition. The actual department outputs illustrate the disconnect:

Legal Dept: “We recommend we take a *cautiously supportive but concerned* position [...] We should endorse the SEC’s goal of closing regulatory gaps that allow exchange-like functions to operate without appropriate oversight.”

Business Strategy: “We should take a *cautiously supportive* position [...] Community banks face unfair competition from unregulated crypto/DeFi platforms [...] Broader exchange definition helps ensure all securities trading platforms operate under similar regulatory frameworks.”

Gov Affairs: “*Support with Strategic Conditions* [...] Frame this as essential for fair competition: community banks have

long operated under comprehensive regulatory oversight while competing against unregulated platforms.”

Synthesis (General Counsel): “ICBA *strongly opposes* [...] the proposed rule would create a disproportionate regulatory burden on community banks [...] community banks would face the same complex compliance requirements as large multinational institutions.”

The synthesis agent amplified a secondary concern (compliance costs) into a primary position, overriding the organization’s core institutional identity. The synthesis step, designed for strategic optimization, instead produced strategic *miscalculation*—the departments’ shared emphasis on competitive fairness and level-playing-field benefits was discarded in favor of a regulatory burden frame that contradicts ICBA’s actual filed position.

4.5 Deliberation Value-Add

Despite the ICBA error, Condition C produced qualitatively richer outputs. The Coinbase deliberative comment integrated arguments about statutory authority (legal), market competitiveness (business), and Congressional engagement strategy (government affairs) into a more layered document than its monolithic counterpart. The internal department outputs also provide interpretable traces of organizational reasoning—a form of “chain-of-thought” at the institutional level that a human analyst can inspect. A regulatory attorney could examine how the simulated legal department frames its strongest statutory argument, or how the government affairs team anticipates coalition dynamics—providing actionable intelligence for drafting preamble language that preemptively addresses likely objections.

4.6 Forward-Looking Application: Simulating Comments on an Anticipated Rule

The main experiment validates against a known comment record but raises contamination concerns. To demonstrate the tool’s intended use case—previewing a comment landscape before it forms—we apply the same architectures to an anticipated SEC rulemaking where no public comment record exists.

In July 2025, SEC Chair Paul Atkins announced “Project Crypto,” a comprehensive initiative to create a formal regulatory framework for crypto asset issuance, custody, and trading. As of February 2026, no formal proposed rule text has been published and no comment period has opened. We constructed a neutral summary of the anticipated provisions based on public statements by SEC leadership and applied Conditions B and C to the same eight organizations.

The predicted landscape is strikingly different from Rule 3b-16: seven of eight organizations support the anticipated framework under both conditions, reflecting the Atkins SEC’s more industry-friendly posture. Two position shifts occurred between conditions: Consensys intensified from “somewhat” to “strongly” support under deliberation, and Uniswap Labs shifted from opposition (Condition B) to support (Condition C)—the deliberative architecture’s departmental analysis apparently resolved a tension between Uniswap’s DeFi-skepticism-of-regulation default and the framework’s proposed safe harbors for decentralized protocols. No ICBA-type flip occurred. These outputs cannot be validated, but they illustrate how a regulatory agency could use organizational

Table 4: Predicted positions on anticipated “Project Crypto” rulemaking (no ground truth available).

Organization	Cond B	Cond C
Coinbase	Strongly sup	Strongly sup
Consensus	Somewhat sup	Strongly sup
Uniswap Labs	Somewhat opp	Somewhat sup
SIFMA	Somewhat sup	Somewhat sup
Better Markets	Somewhat sup	Somewhat sup
ICBA	Somewhat sup	Somewhat sup
Chamber of Comm.	Somewhat sup	Somewhat sup
Blockchain Assn.	Somewhat sup	Somewhat sup

simulations to anticipate the comment landscape before publishing a proposed rule.

5 Discussion

The deliberative architecture trades *robustness* for *richness*. Monolithic agents directly encode institutional identity and achieve perfect position prediction; deliberative agents produce more nuanced, multi-dimensional reasoning but introduce fragility when the synthesis step must reconcile departmental tensions. We explore implications below.

5.1 Implications for Regulatory Design

Just as AI-mediated deliberation has shown promise in helping citizens find common ground on policy questions [9], organizational agent simulations suggest a direction worth exploring for policymakers, as the Project Crypto simulation (Section 4.6) concretely illustrates. A regulatory agency drafting a proposed rule could potentially use such simulations to preview likely comment landscapes: which industry groups are likely to oppose, on what legal or economic grounds, and with what rhetorical framing? The deliberative architecture is particularly valuable here because its department-level outputs function as *explainable AI for policy simulation*—a regulator can inspect not just what an organization is predicted to argue, but trace the reasoning through legal risk analysis, business impact assessment, and political strategy. This interpretability aligns with HCI principles of transparency and user control: the human analyst remains in the loop, using simulated outputs as a starting point for their own judgment rather than as a black-box prediction. Significant further validation—across multiple rulemakings, models, and organizational samples—would be needed before such tools could responsibly inform regulatory design.

5.2 The Training Data Circularity Problem

The most significant methodological concern is that our underlying model (Claude Sonnet) almost certainly encountered these organizations’ actual SEC comment letters during pretraining—they are public filings indexed by search engines. This raises the question of whether the model is *reasoning* about what these organizations would argue or merely *retrieving* what they actually argued.

We take this concern seriously but note three mitigating factors. First, our experiment is a *controlled comparison* across three architectures using the same model: even if absolute fidelity is inflated by training data memorization, the *relative* differences between conditions remain informative. The fact that Condition B perfectly reproduces positions while Condition C introduces an error tells us something real about how the deliberative architecture transforms organizational reasoning, regardless of the model’s prior knowledge. Second, the ICBA failure in Condition C is itself evidence against pure retrieval—if the model were simply looking up ICBA’s filed position, it would not have flipped to opposition. The deliberation process generated a novel (and incorrect) output, demonstrating that the architecture actively reshapes the model’s reasoning rather than passing through memorized content.

The current Rule 3b-16 experiment tests whether architectural choices reshape organizational reasoning even when the model has prior exposure, a question the ICBA result answers affirmatively—the deliberative architecture overrode what the model likely “knows” about ICBA’s actual position. This is a feature of the study design, not just a limitation. The Project Crypto simulation (Section 4.6) complements this by answering a different question: can these architectures generate plausible organizational positions on something the model has never seen commented on? Together, the two experiments address different facets of the contamination concern—the first demonstrates that architectural effects are real regardless of prior knowledge, the second demonstrates the tool’s intended predictive use case, which eliminates contamination by design but forfeits validation.

There is also a recursive dimension to this work: we use AI to simulate comments on rules that regulate technology, at a moment when regulatory comment processes are themselves increasingly populated by AI-generated submissions—the very problem the Comment Integrity and Management Act [10] was designed to address.

5.3 Limitations

Beyond the circularity concern, the sample of eight organizations is small, and our argumentative range metric relies on keyword detection rather than expert evaluation of argument quality. The single-model design means we cannot distinguish model-specific behavior from generalizable architectural effects. Future work should use different models for generation and evaluation, expand the organizational sample, recruit domain experts for qualitative assessment, and test across multiple rulemakings with varying political dynamics.

5.4 Workshop Questions

We pose four questions tailored to our findings:

- (1) *Should the synthesis step have access to the organization’s mission statement as a binding constraint?* The ICBA failure suggests that without an explicit institutional identity anchor, the synthesis agent optimizes for strategic plausibility over organizational fidelity. What is the right balance between creative strategic reasoning and mission-constrained argumentation?
- (2) *Are position accuracy and argument richness fundamentally in tension?* Our results show a tradeoff: the simpler architecture

gets positions right while the richer architecture produces better arguments but occasionally wrong positions. Is this an artifact of our design or a deeper property of multi-agent deliberation?

- (3) *How should we evaluate strategic reasoning beyond position prediction?* An agent might predict the correct position for the wrong reasons. What evaluation frameworks—drawn from legal analysis, rhetoric, or argumentation theory—could assess whether simulated arguments would actually be *persuasive* to the regulatory audience?
- (4) *What design patterns make organizational simulations useful for human decision-makers?* Our deliberative architecture produces department-level traces that a regulatory attorney could inspect. What HCI principles should guide the design of interfaces that present these simulated organizational perspectives to policymakers?

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A LLM Disclosure

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